

CAD Mechanical Design Portfolio Report

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Developed Using

Fusion360

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1 INTRODUCTION

Computer-Aided Design (CAD) is one of the most important tools used in modern Mechanical Engineering. CAD software enables engineers to create precise three-dimensional models, generate engineering drawings, visualize assemblies, and communicate designs effectively.

Throughout this portfolio, Autodesk Fusion 360 was used to design a variety of mechanical components and systems. The projects focused on mechanical design fundamentals, power transmission elements, robotics-related structures, and technical documentation.

The objective of these projects was to develop practical CAD skills while understanding the engineering principles behind each design.

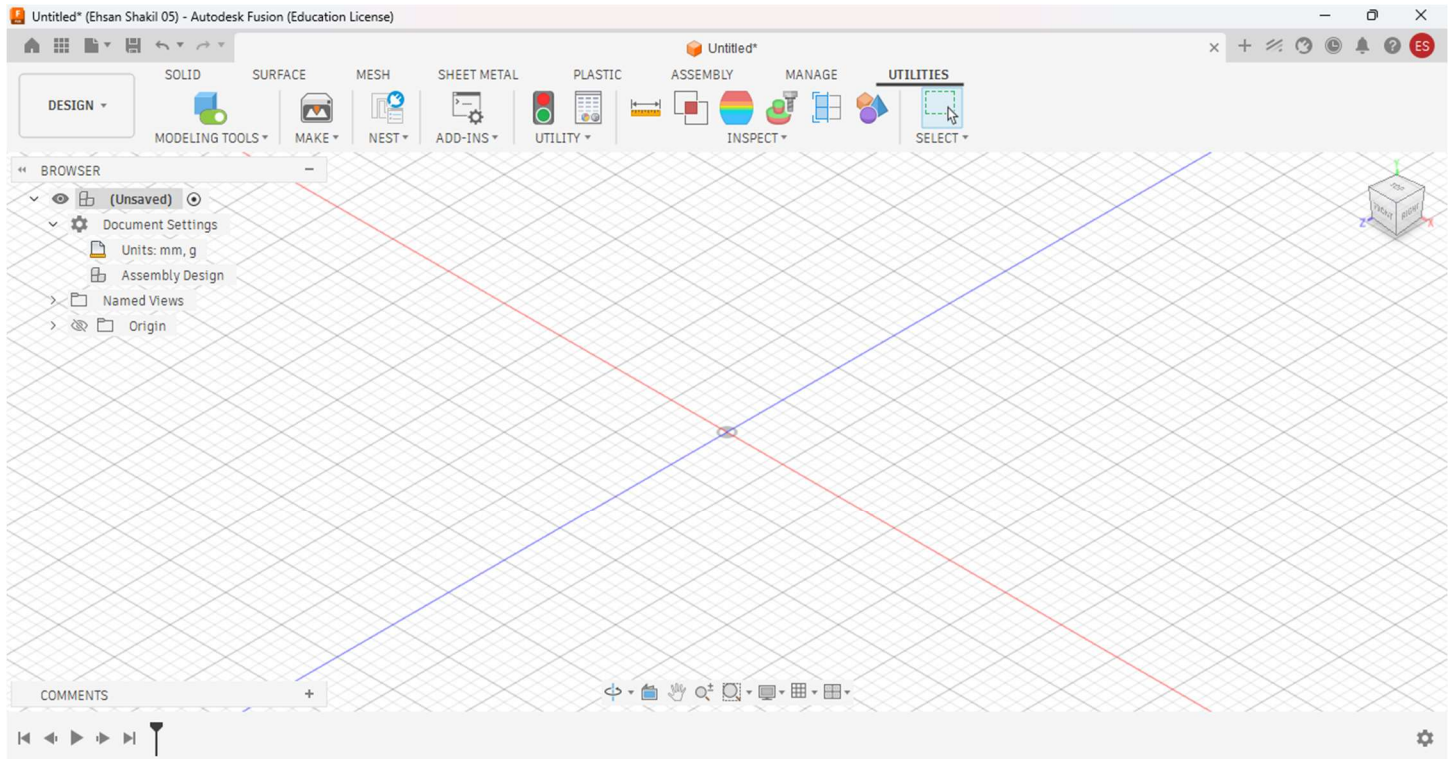


Figure 1: Fusion 360 Workspace Screenshot

2 BEARING MOUNT DESIGN

2.1 PROJECT OVERVIEW

The Bearing Mount project introduced the fundamentals of mechanical CAD design. The component was designed to support a rotating shaft through the use of a bearing while providing mounting features for attachment to a supporting structure.

2.2 DESIGN FEATURES

- Parametric sketching
- Extruded base geometry
- Central bearing housing
- Mounting holes
- Engineering drawing generation

2.3 ENGINEERING CONSIDERATIONS

The bearing mount must provide adequate support and alignment for the shaft while maintaining structural rigidity. Proper hole placement and dimensional accuracy are important to ensure reliable assembly and operation.

2.4 SKILLS APPLIED

- Sketch creation
- Dimensional constraints
- Extrusions
- Hole features
- Technical drawings



Figure 2: Bearing Mount Render

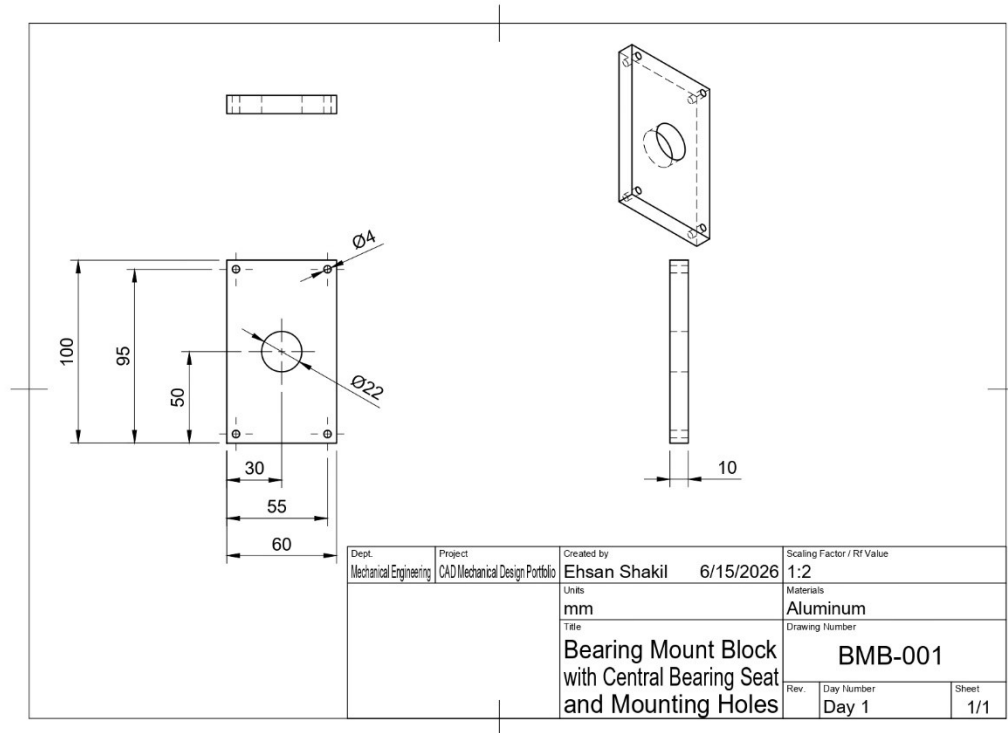


Figure 3: Bearing Mount Engineering Drawing

3 SHAFT COUPLING DESIGN

3.1 PROJECT OVERVIEW

The Shaft Coupling project focused on the design of a mechanical component used to connect two shafts and transmit rotational power. The coupling was modeled using circular features, bolt patterns, and central shaft bores.

3.2 DESIGN FEATURES

- Circular geometry
- Shaft bore
- Bolt circle pattern
- Mechanical fastening features
- Symmetrical design

3.3 ENGINEERING CONSIDERATIONS

Shaft couplings are widely used in motors, pumps, compressors, and industrial machinery. The design must ensure efficient torque transmission while maintaining shaft alignment.

The component was modeled using Aluminum 6061 and rendered with a satin finish to represent a realistic manufactured appearance.

3.4 SKILLS APPLIED

- Pattern features
- Precision dimensioning
- Mechanical component design



Figure 4: Shaft Coupling Render

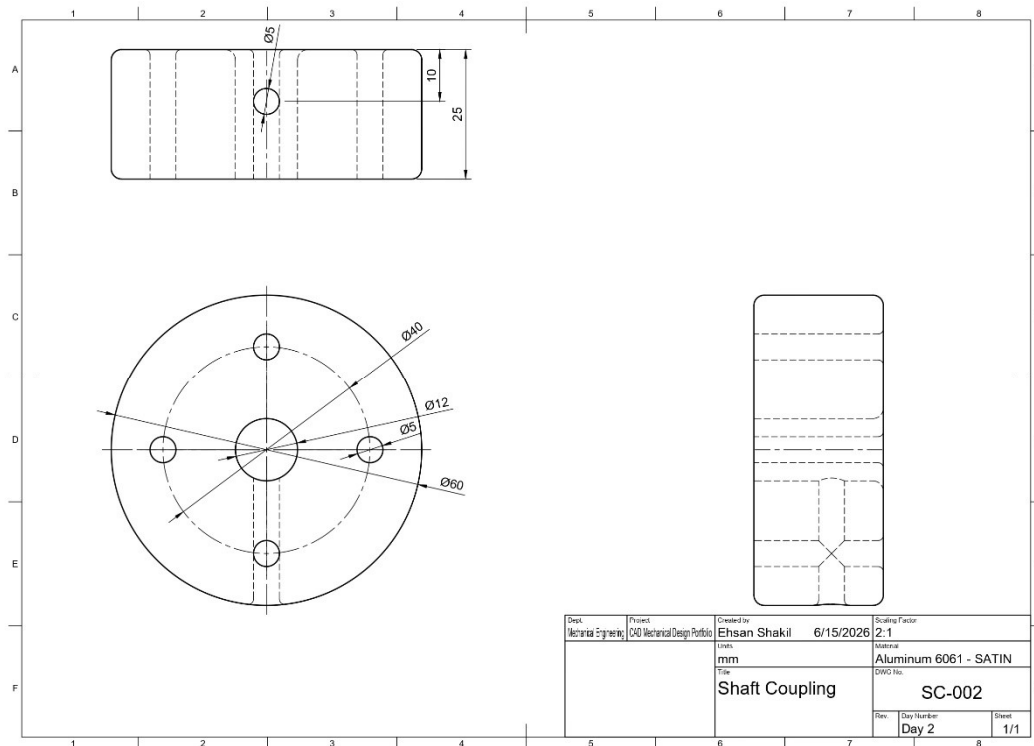


Figure 5: Shaft Coupling Engineering Drawing

4 SPUR GEAR & GEAR TRAIN DESIGN

4.1 PROJECT OVERVIEW

This project involved the creation of a spur gear pair and a simple gear train assembly. The design demonstrated motion transfer between rotating components and introduced the concept of gear ratios.

4.2 DESIGN FEATURES

- Spur gear geometry
- Circular tooth pattern
- Revolute joints
- Gear train assembly

- Motion relationship visualization

4.3 ENGINEERING CONSIDERATIONS

The gear train consisted of a 20-tooth gear driving a 40-tooth gear. This produced a gear ratio of 2:1, reducing rotational speed while increasing output torque.

Gear systems are essential components in robotics, automotive systems, manufacturing equipment, and industrial machinery.

4.4 SKILLS APPLIED

- Parametric design
- Circular patterns
- Assemblies
- Motion relationships
- Mechanical power transmission concepts



Figure 6: Gear Train Assembly

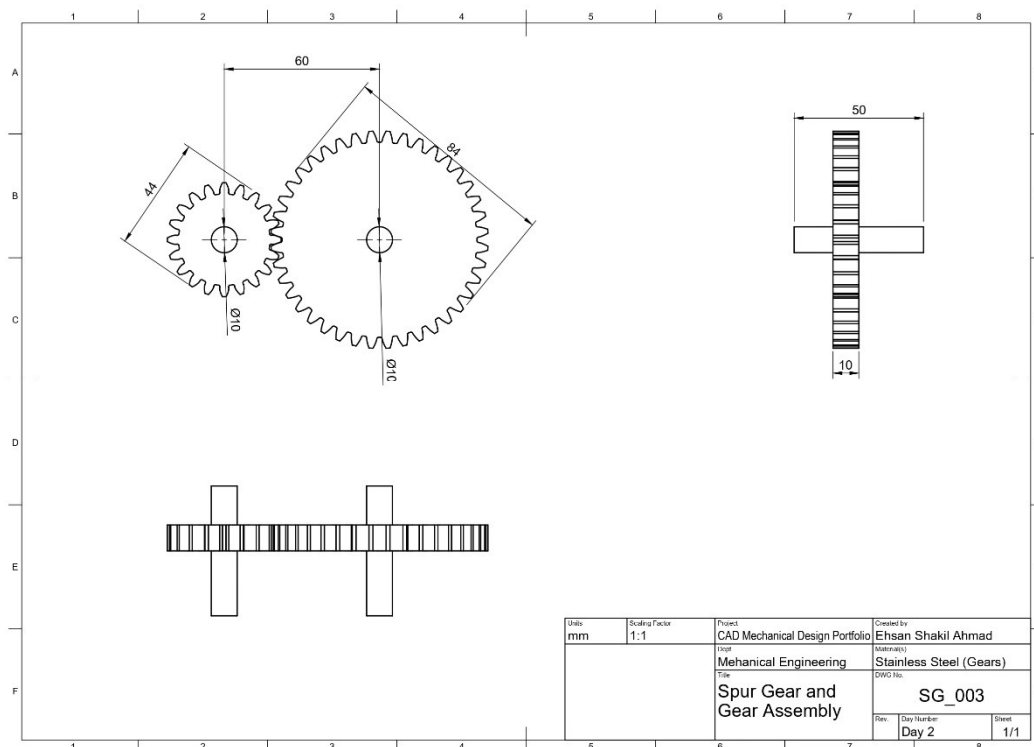


Figure 7: Spur Gear Engineering Drawing

5 ROBOT CHASSIS DESIGN

5.1 PROJECT OVERVIEW

The Robot Chassis project focused on the development of a mechanical platform for a two-wheel mobile robot. The design incorporated mounting locations for electronic and mechanical components commonly used in robotics systems.

5.2 DESIGN FEATURES

- Structural base plate
- Motor mounting locations
- Battery placement area
- Arduino mounting provision
- Motor driver mounting provision
- Caster wheel support

5.3 ENGINEERING CONSIDERATIONS

The objective of the design was to create a compact and functional mobile robot platform while considering component packaging and assembly requirements.

During the design process, several alignment and packaging issues were identified. The caster wheel mount, wheel placement, and mounting locations for the Arduino, motor driver, and battery holder require refinement in future iterations. Despite these limitations, the project successfully demonstrated the complete design workflow and introduced important concepts related to robotics-oriented mechanical design.

5.4 SKILLS APPLIED

- Mechanical layout design
- Design for assembly
- Component packaging
- Robotics-focused CAD modeling

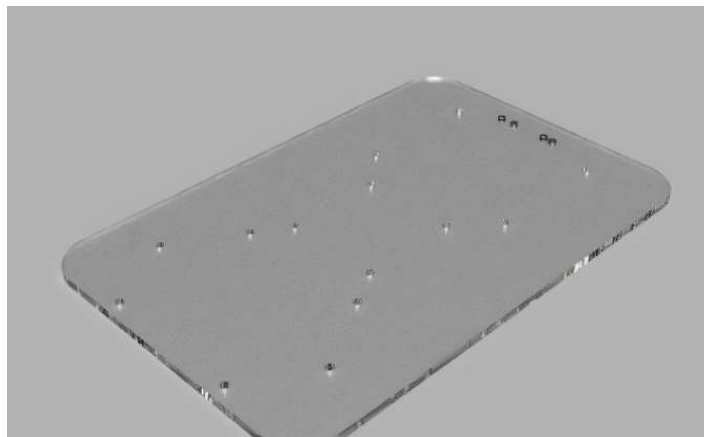


Figure 8: Robot Chassis Render

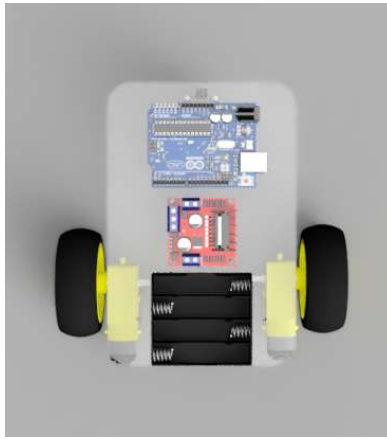


Figure 9: Chassis with Components

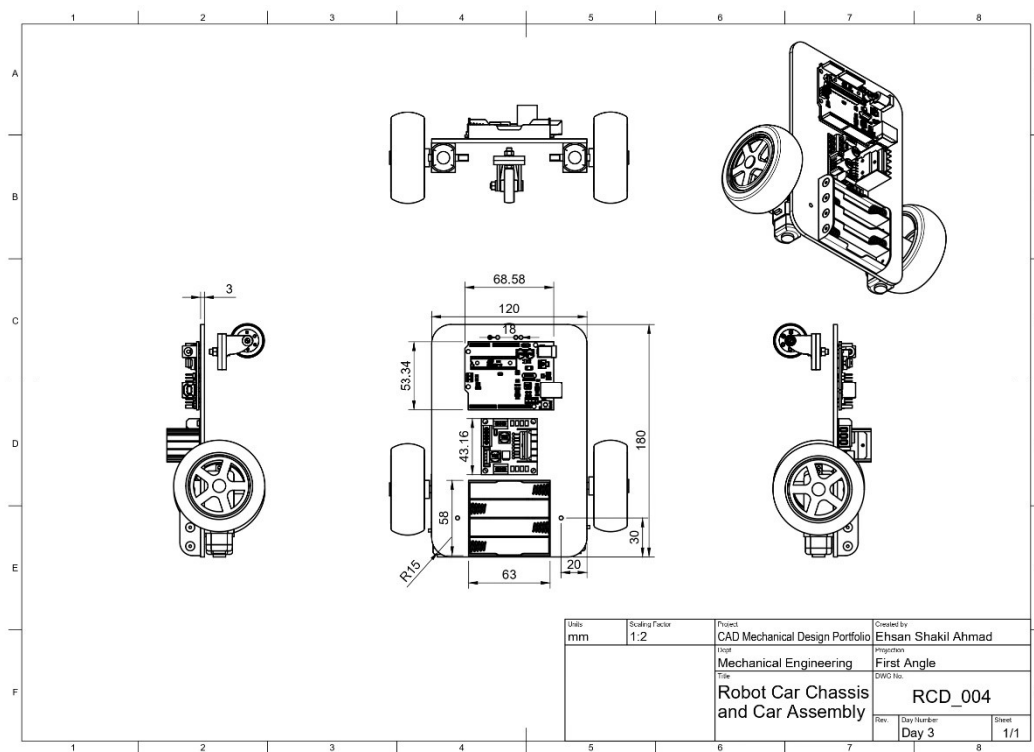


Figure 10: Robot Engineering Drawing

6 SKILLS DEVELOPED

The completion of these projects significantly improved practical CAD and engineering design skills.

Skills developed include:

- Sketching
- Constraints
- Parametric Modeling
- Extrusions
- Revolved Features
- Hole Features

- Circular Patterns
- Assemblies
- Engineering Drawings
- Rendering
- Technical Documentation

These skills form a strong foundation for future mechanical and robotics-related design projects.

7 CHALLENGES FACED

Several challenges were encountered during the completion of the projects:

- Learning the Autodesk Fusion 360 interface
- Understanding parametric design concepts
- Fully constraining sketches
- Creating accurate hole patterns
- Generating professional engineering drawings
- Understanding assembly relationships and joints
- Designing components with practical manufacturing considerations

Overcoming these challenges improved both technical understanding and confidence in CAD modeling.

8 FUTURE IMPROVEMENTS

Future work will focus on expanding both mechanical design complexity and robotics integration.

Potential future projects include:

- Complete Mobile Robot Assembly
- Motion Simulation Studies
- Self-Balancing Robot CAD
- CNC Plotter Design
- 3D Printable Mechanical Components
- Advanced Robotics Assemblies
- Finite Element Analysis (FEA)
- Manufacturing-Oriented Design Projects

These projects will further strengthen mechanical design and robotics engineering skills.

9 CONCLUSION

This CAD portfolio demonstrates the successful completion of four mechanical design projects using Autodesk Fusion 360. Through these projects, valuable experience was gained in parametric modeling, engineering drawings, mechanical component design, and robotics-related CAD development.

The portfolio provides a solid foundation for future work in mechanical engineering, robotics, mechatronics, and product development. The skills acquired during these projects will support more advanced design, simulation, and engineering challenges in the future.